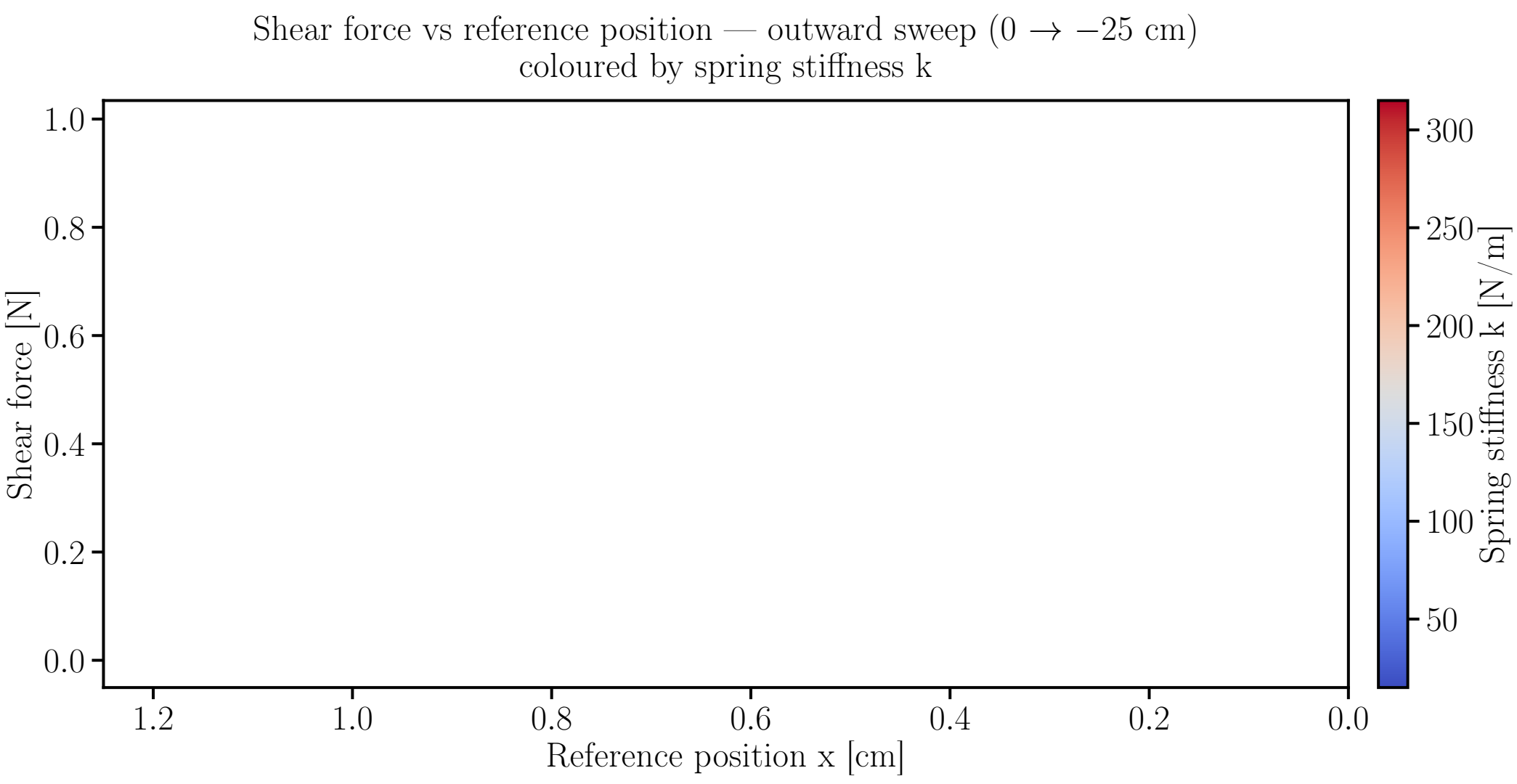

Shear force vs reference position — outward sweep (0 → −25 cm)
coloured by spring stiffness k
Shear force [N]
1.0
0.8
0.6
0.4
0.2
0.0
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Reference position x [cm]
Spring stiffness k [N/m]
300
250
200
150
100
50